

10. Data pipelines

10.1. Definition & differentiation

Lesson objectives

Data pipelines: Definition & differentiation

1. Define a data pipeline and its typical use case.
2. Differentiate between ETL and ELT.
3. Explain how Change Data Capture (CDC) impacts data pipelines.
4. Explain how Slowly Changing Dimensions (SCD) impact data pipelines.

Data pipeline definition

IBM defines a data pipeline as:

A data pipeline is a method in which raw data is ingested from various data sources and then ported to data store, like a data lake or data warehouse, for analysis.

Before data flows into a data repository, it usually undergoes some data processing.

This is inclusive of data transformations, such as filtering, masking, and aggregations, which ensure appropriate data integration and standardization.

<https://www.ibm.com/topics/data-pipeline>

Isn't that ETL, I mean ELT?

Extract-Transform-Load (ETL)

Popular solution for decades:

- **Extract** data (usually from a database)
- **Transform** the data to prepare it for analytics
- **Load** the data in a data warehouse

Extract-Load-Transform (ELT)

Data lakes encourage us to rethink the order:

- **Extract** data (from a variety of data sources in batch or streaming modes)
- **Load** the data to the data lake
- **Transform** to prepare for analytics

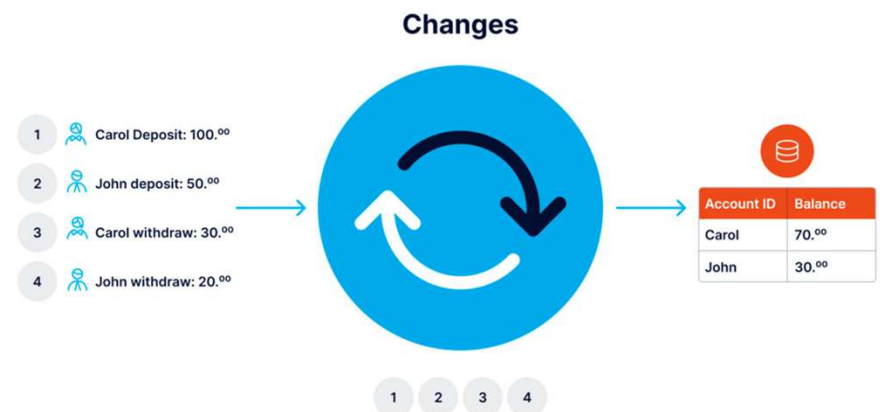
They are both very similar and your data storage architecture will influence your approach.

Data pipelines use ETL/ELT strategies.

Change Data Capture (CDC) defined

Wikipedia defines [CDC](#) as:

In databases, change data capture (CDC) is a set of software design patterns used to determine and track the data that has changed so that action can be taken using the changed data.



Data pipelines that focus on replicating data from mutable data sources are leveraging CDC.

Slowly Changing Dimensions (SCD)

A Slowly Changing Dimension (SCD) is a dimension table that manages & stores both historical & present data over time in a data warehouse/lakehouse.

Multiple approaches exist, but the most popular are:

- **Type 1** – Only maintains current data; no historical data is kept
- **Type 2** – Keeps current as well as historical data in the table made to the row

Data pipelines use SCD strategies when the data being processed is not just simple, insert-only records.

Lesson summary

Data pipelines: Definitions & differentiation

1. A data pipeline ingests raw data, moves it through a number of transformation stages, and makes it available for analysis.
2. The ETL process is the standard approach to managing data in a data pipeline, and consists of extracting it, transforming it, and then loading it.
3. ELT is a different process that extracts data, then loads it, only transforming it when it is needed.
4. Data pipelines may implement CDC and/or SCD strategies.

10. Data pipelines

10.2. Reference architecture

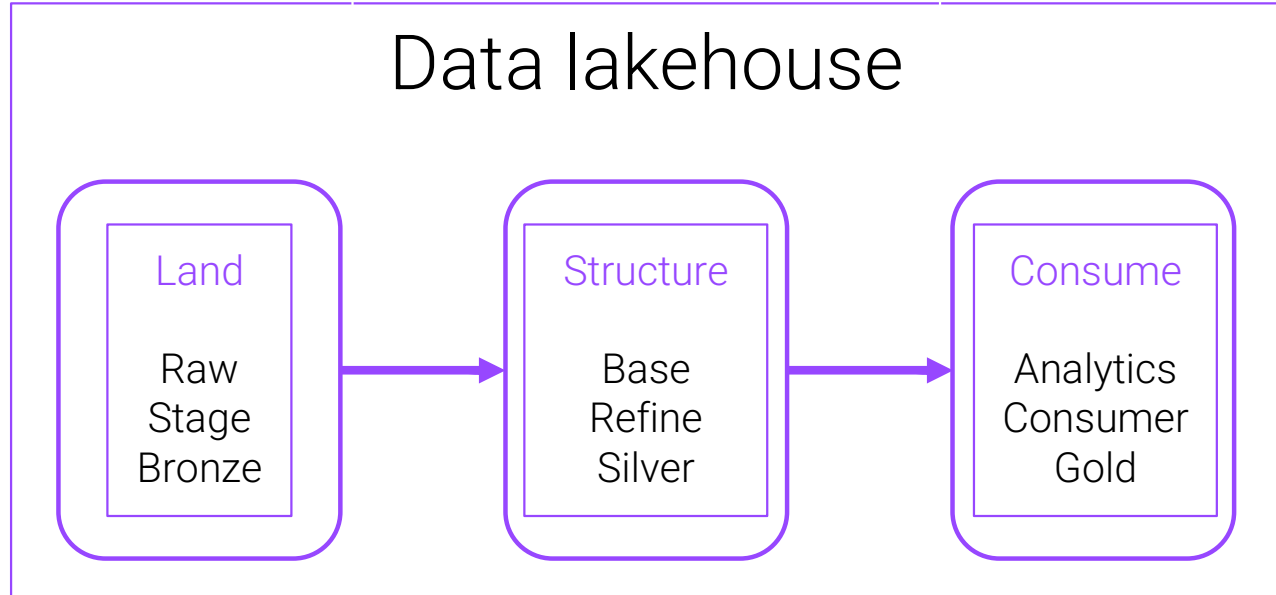
Lesson objectives

Data pipelines: Reference architecture

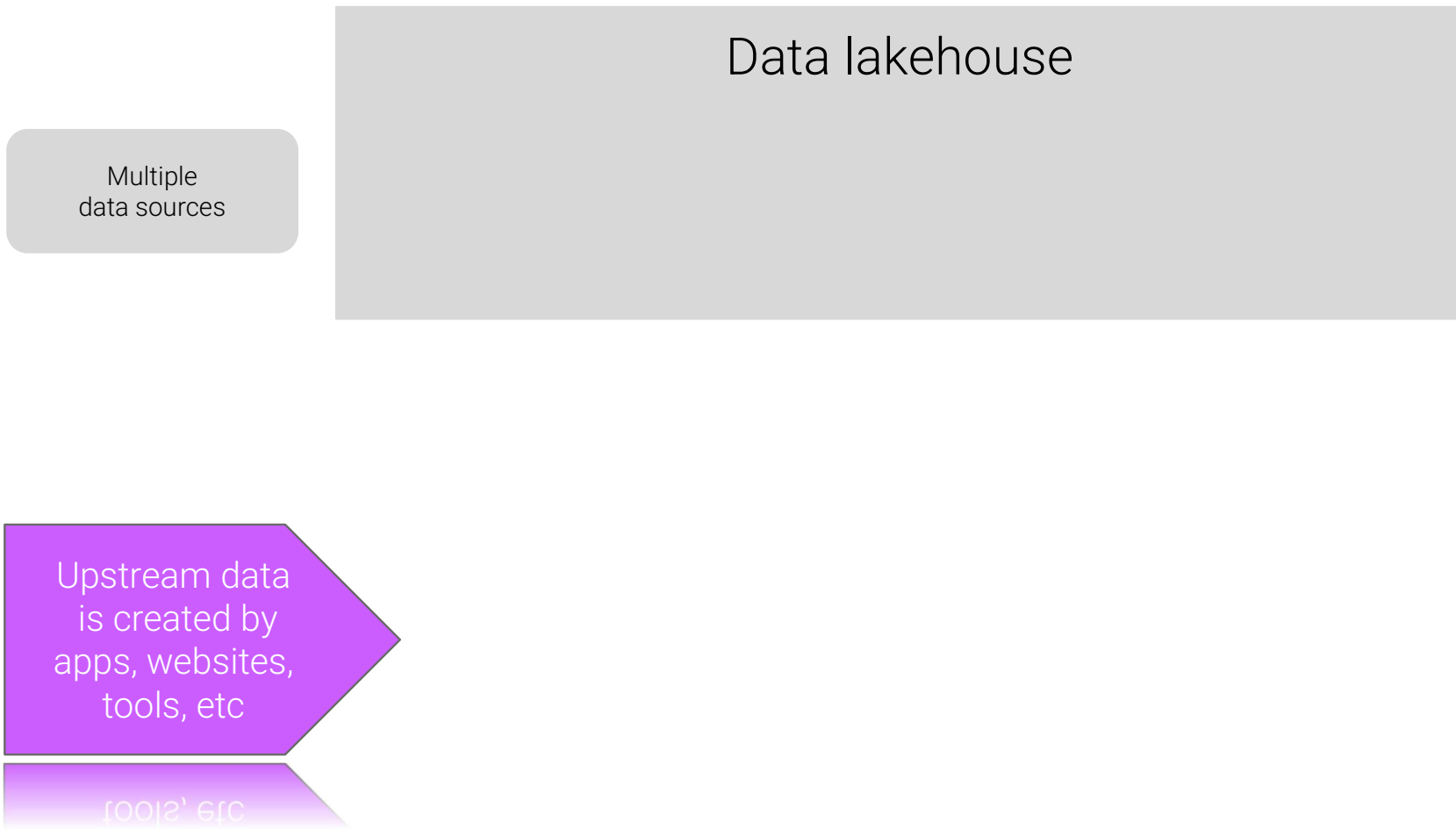
1. Describe the three layers that make up the typical data lakehouse reference architecture.
2. Explain the activities & processes across the architecture.
3. Describe tools & technologies used for the layers and the activities & processes that are executed.

Reference architecture

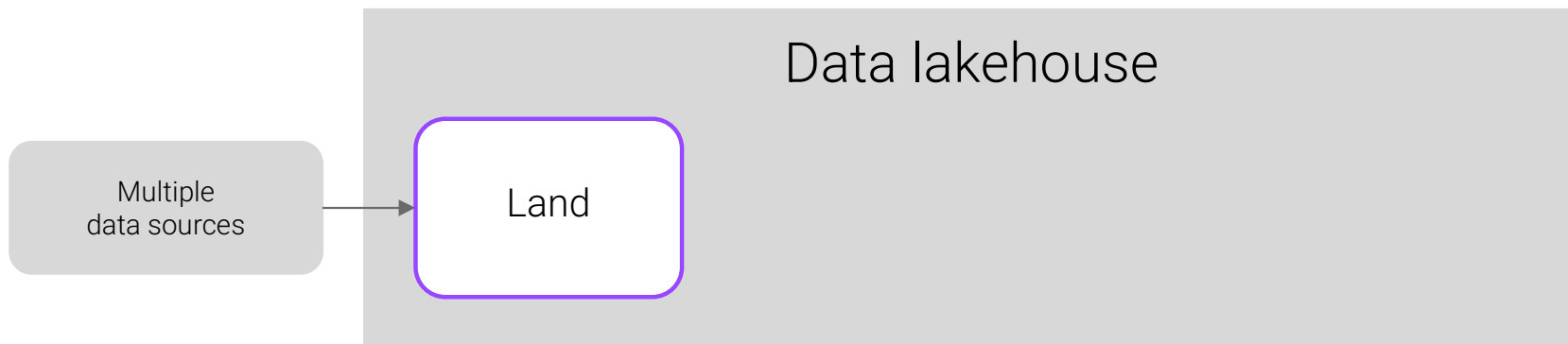
The reference architecture centers around the data lakehouse and how we classify our data assets into distinct zones. Data pipelines populate the zones.



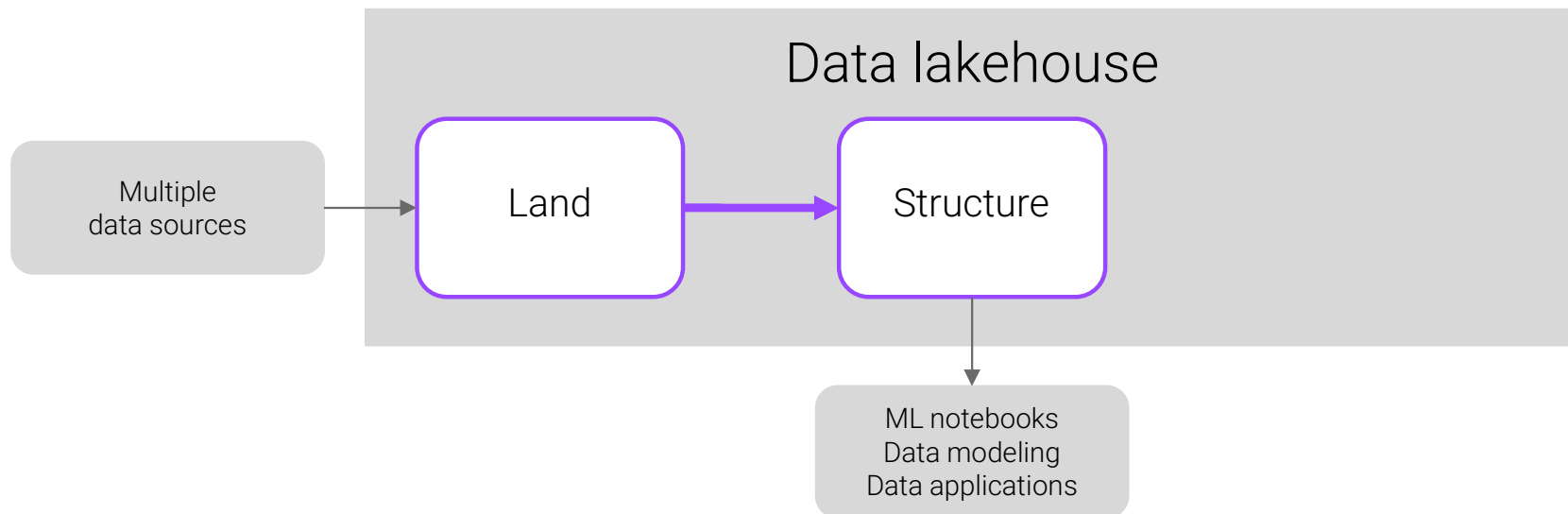
Activities across the architecture



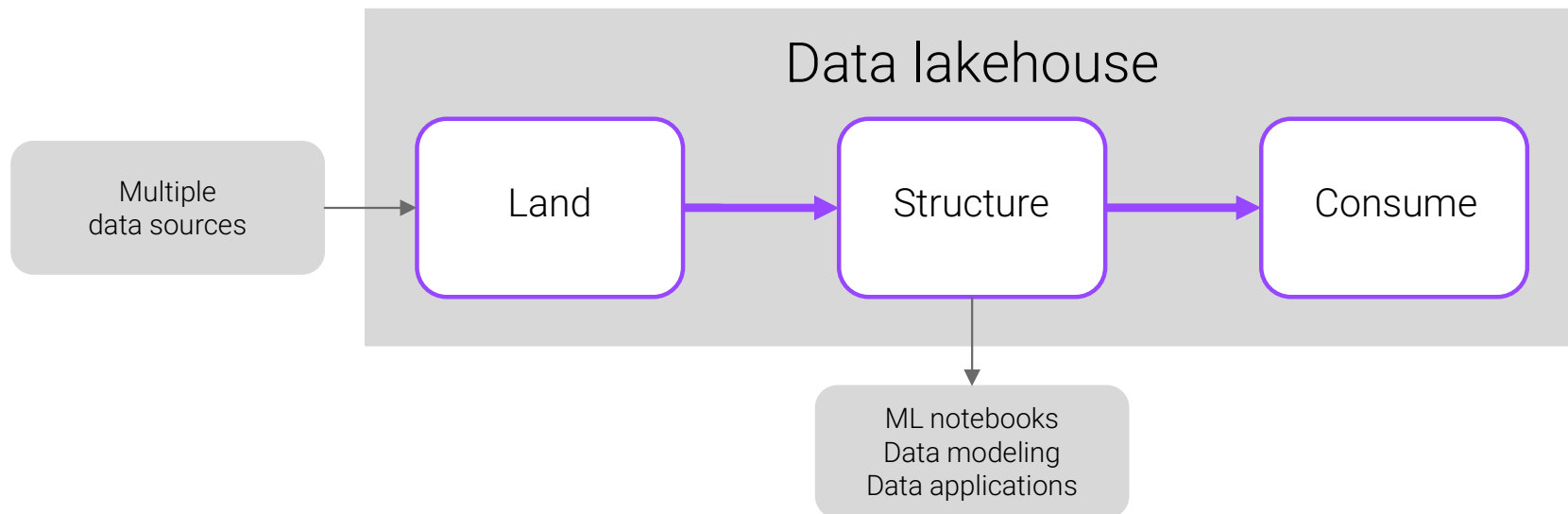
Activities across the architecture



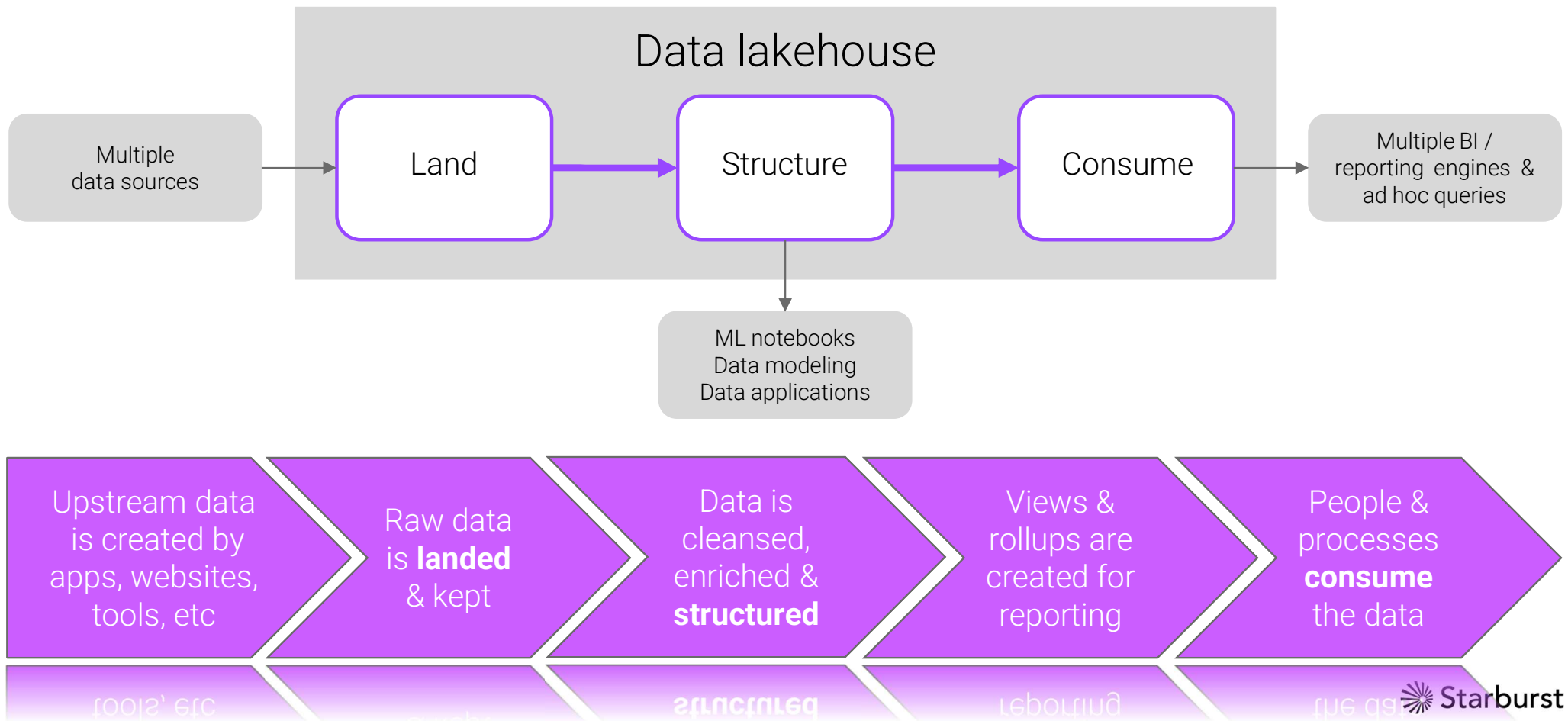
Activities across the architecture



Activities across the architecture



Activities across the architecture



Tools & technologies



Data lakehouse

Tools & technologies

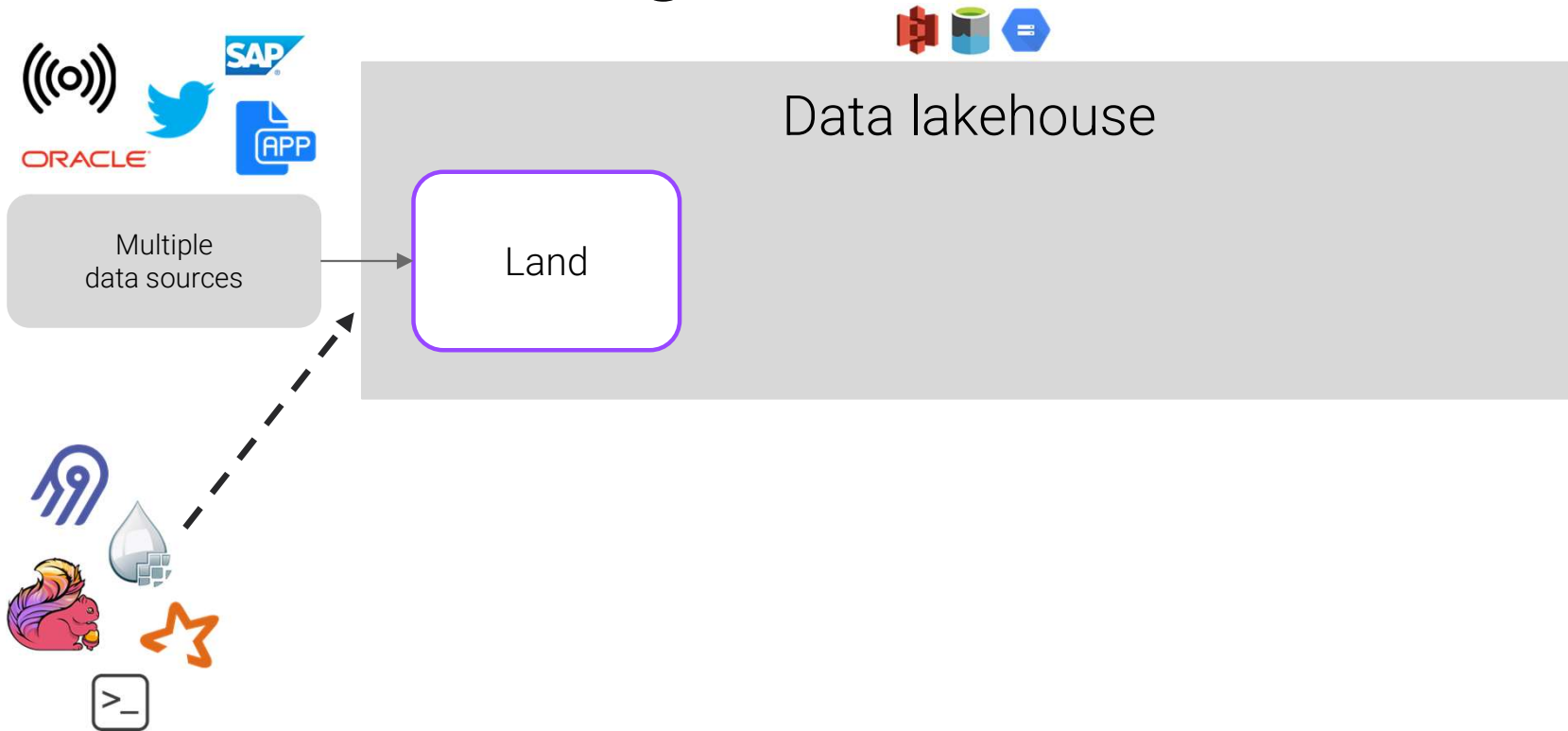


Multiple
data sources

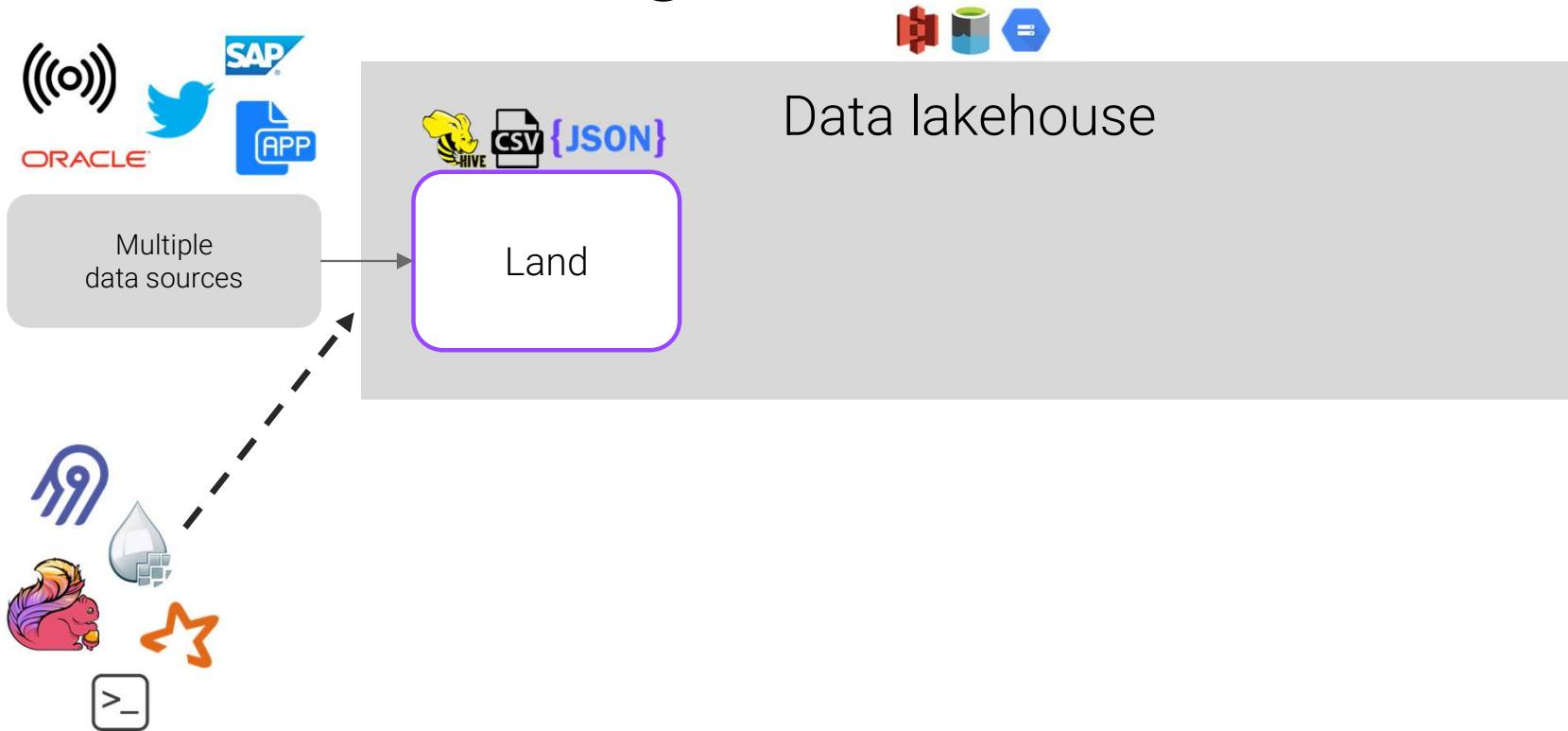


Data lakehouse

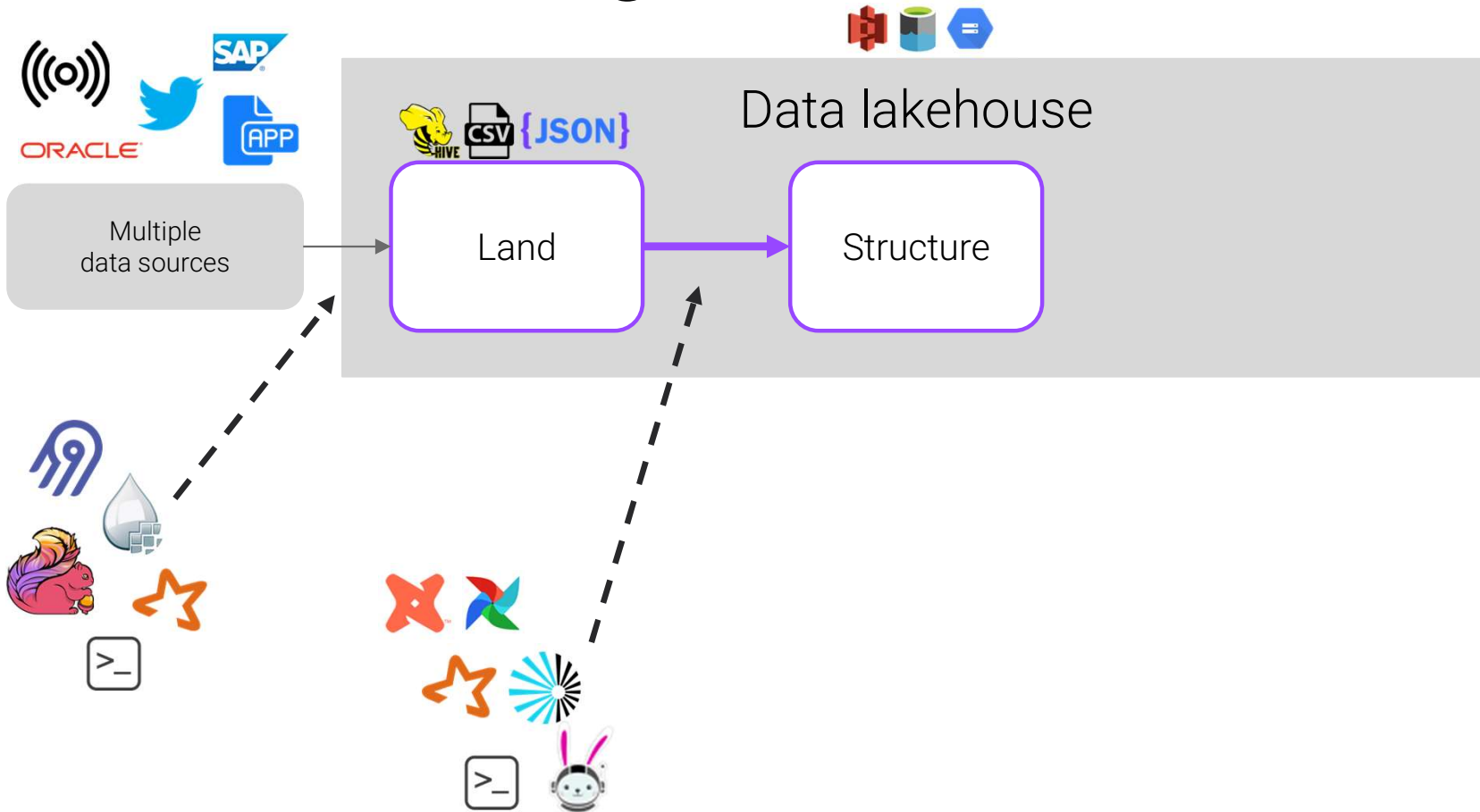
Tools & technologies



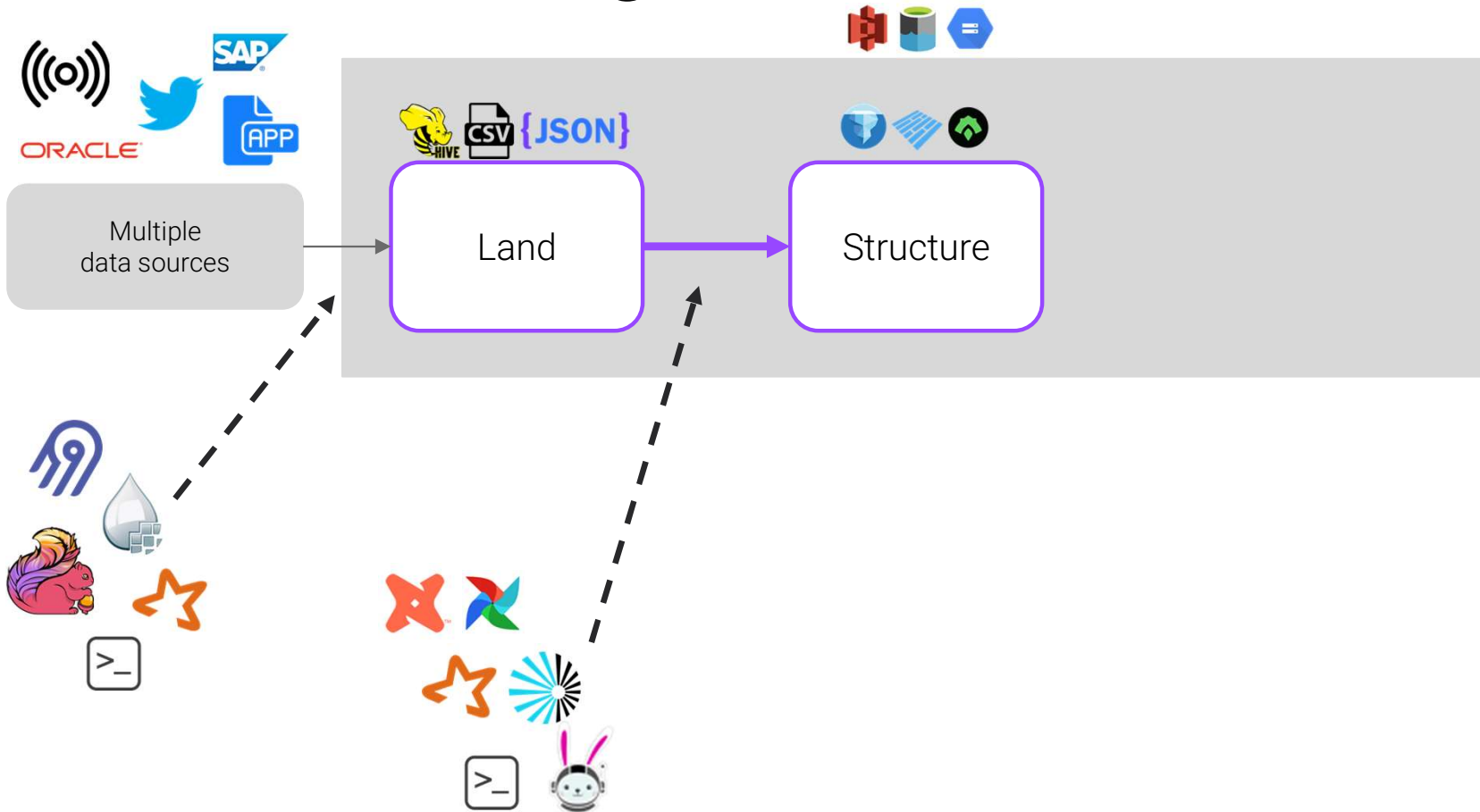
Tools & technologies



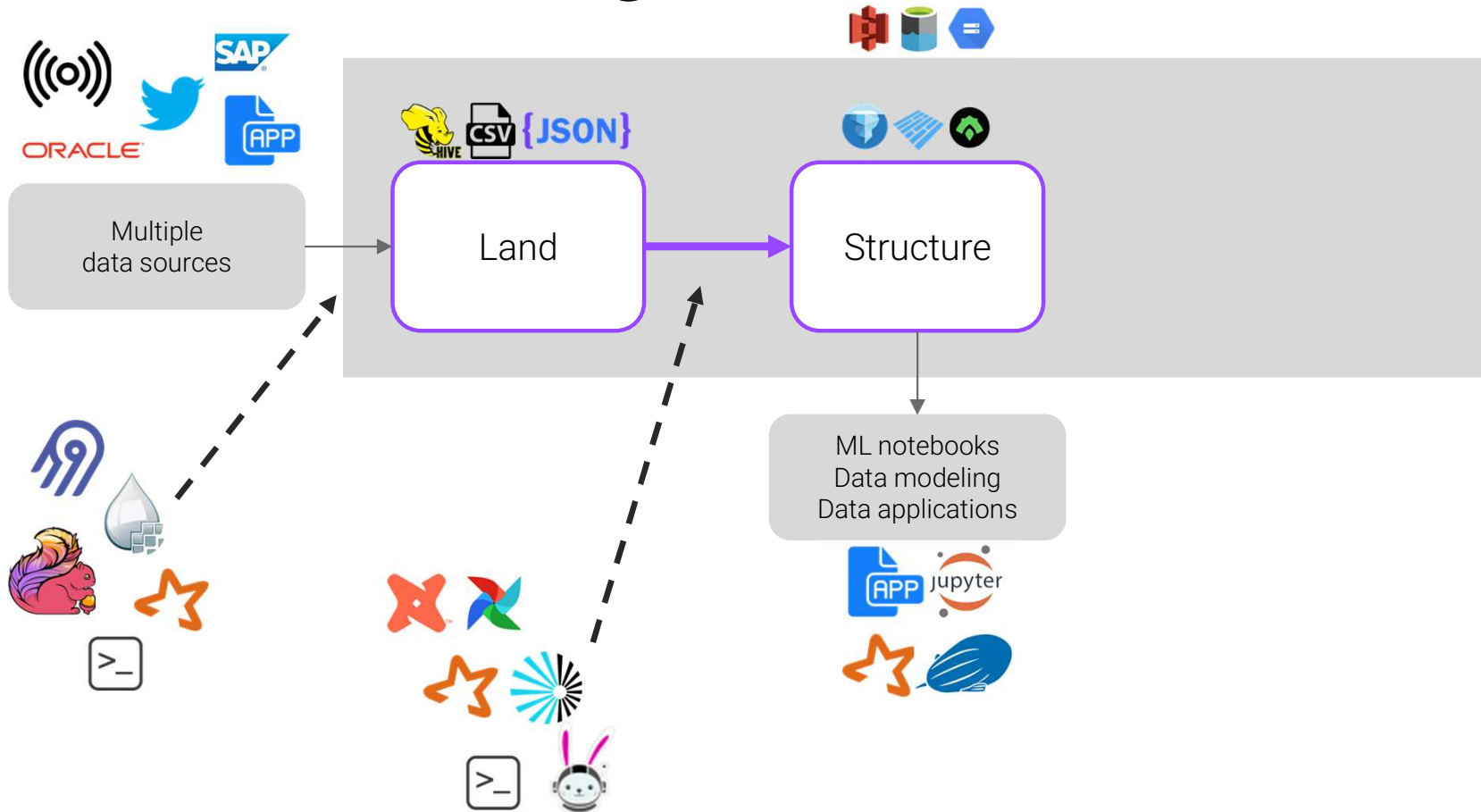
Tools & technologies



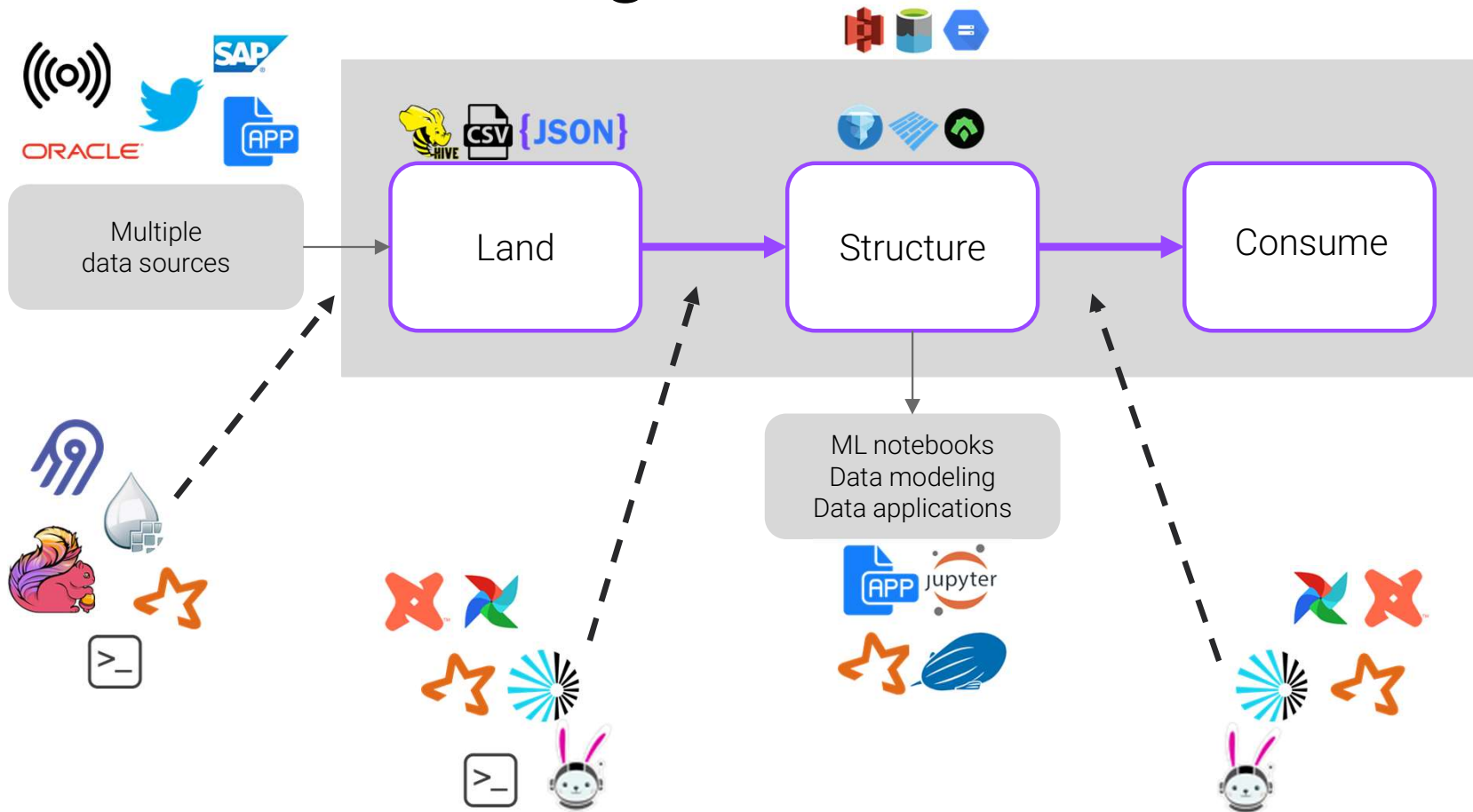
Tools & technologies



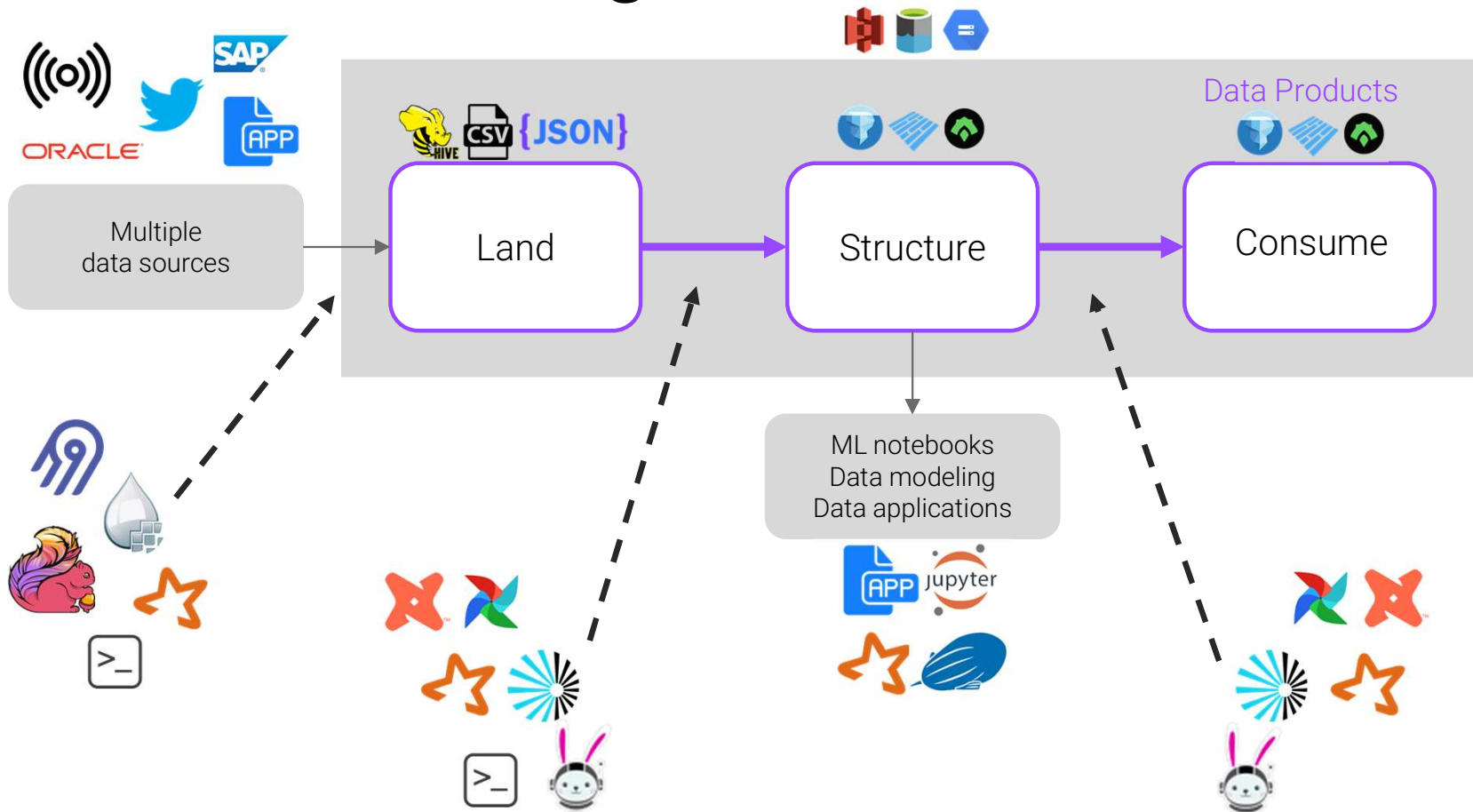
Tools & technologies



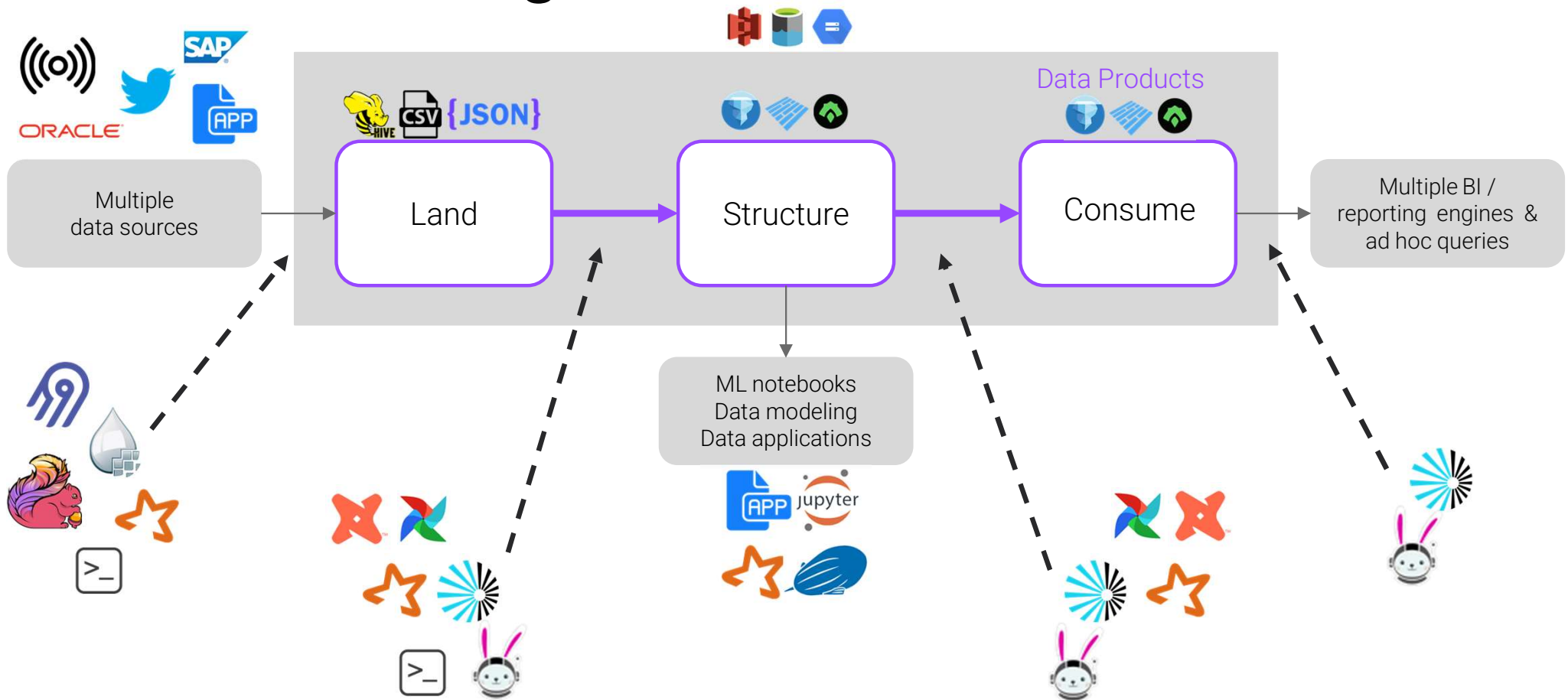
Tools & technologies



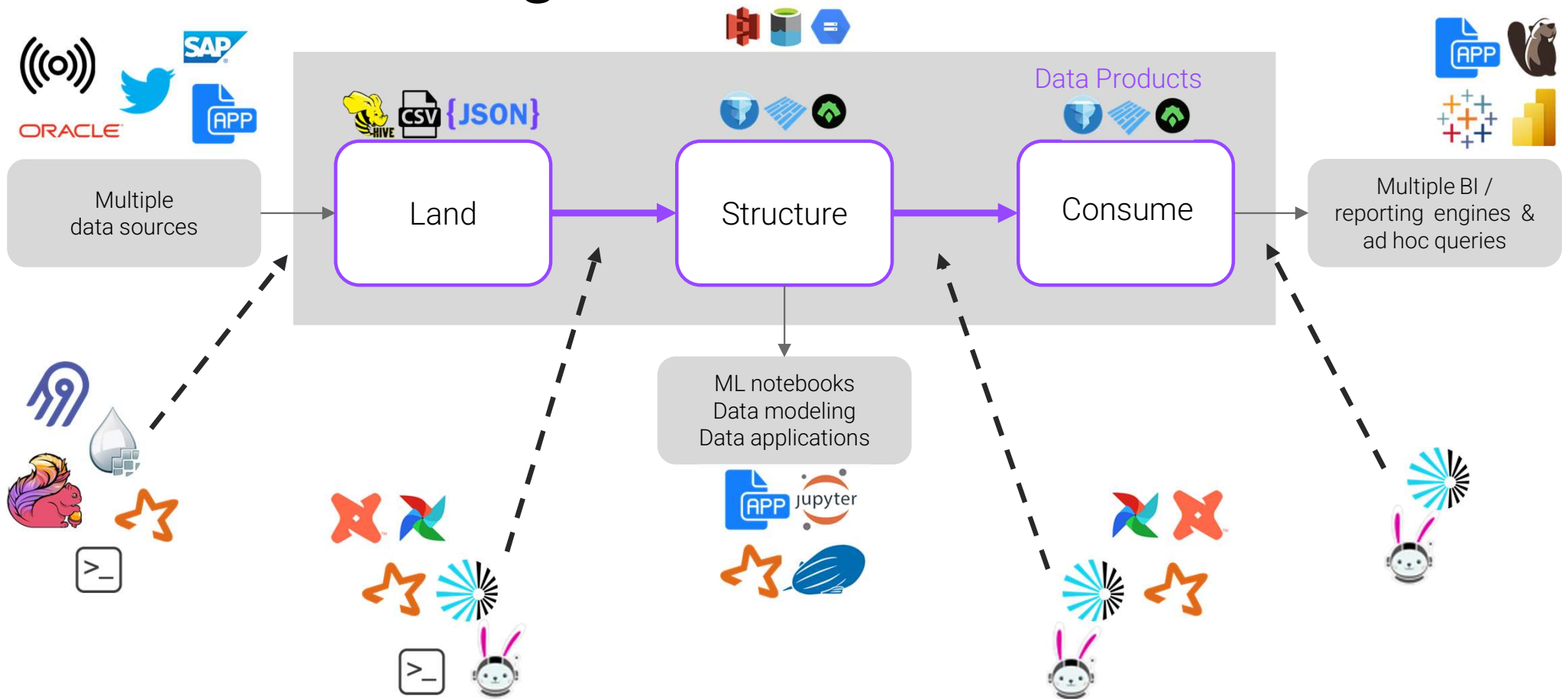
Tools & technologies



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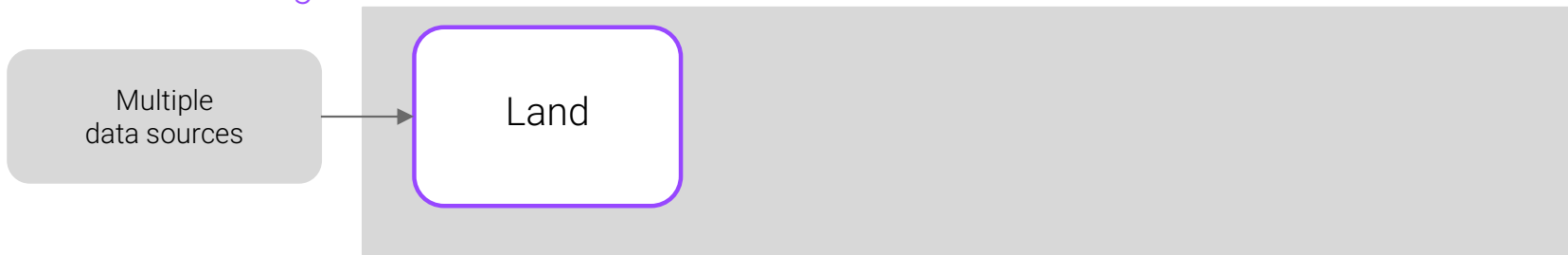
Summary

Ingestion



Summary

Ingestion



Extract

Load

Summary

Ingestion

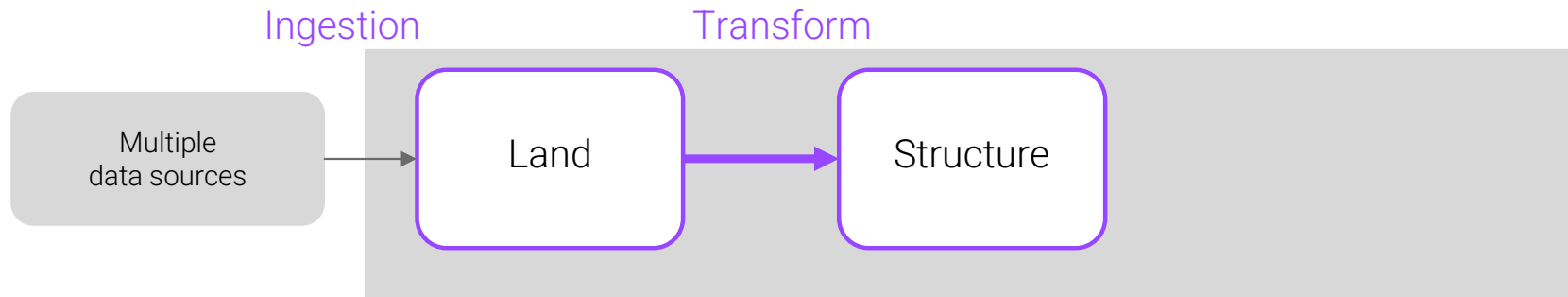


Data pipeline

Extract

Load

Summary



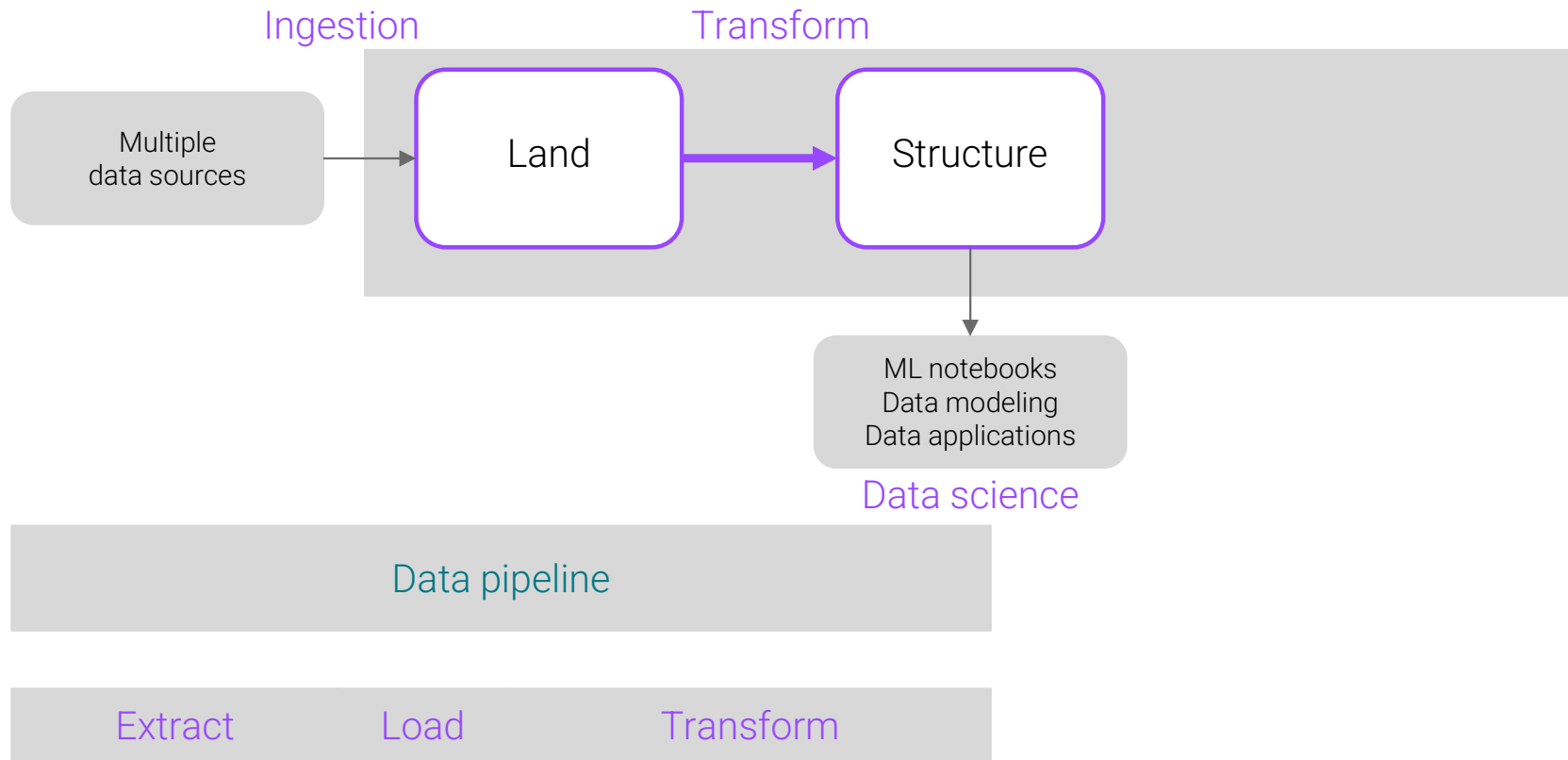
Data pipeline

Extract

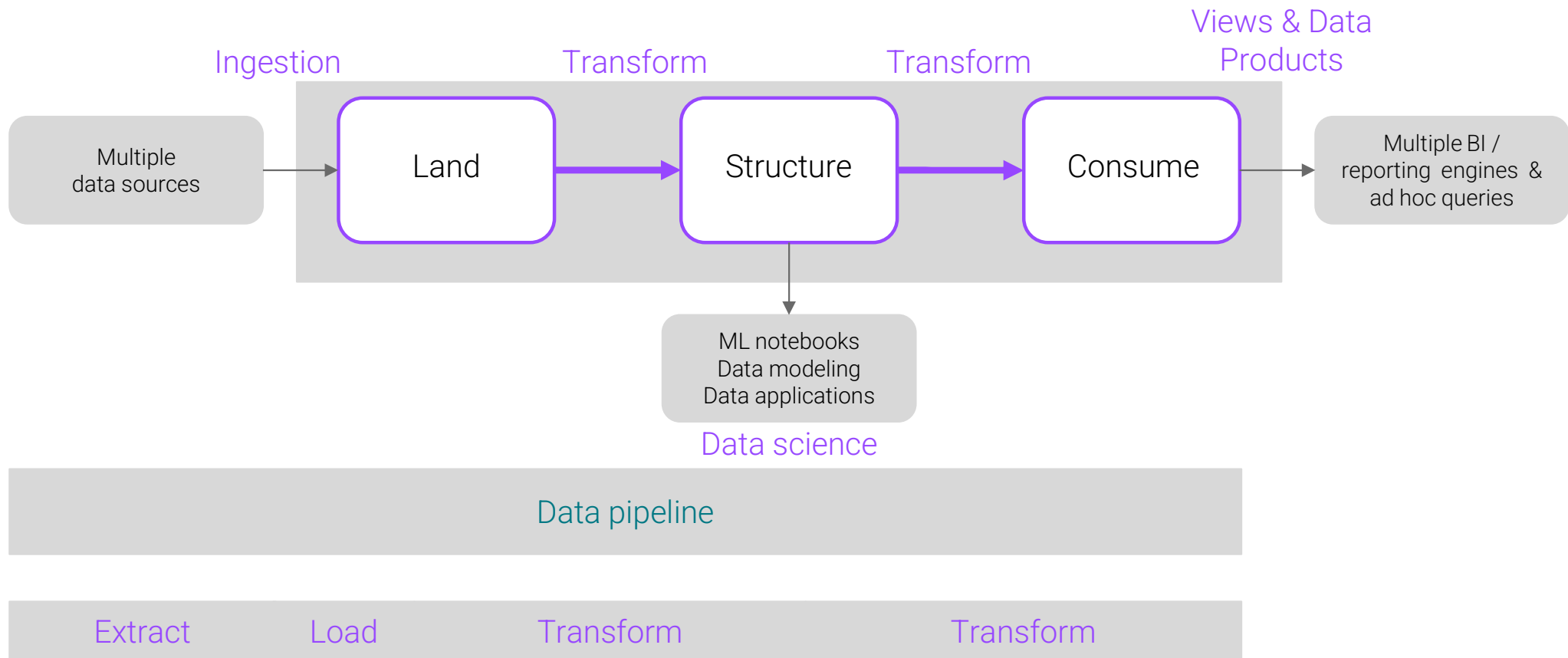
Load

Transform

Summary



Summary



Hands-on exercises

Lab 1: Construct a pipeline with
insert-only transactions (60 mins)

Lab 2: Construct a pipeline with the
MERGE statement (30 mins)

Lesson summary

Data pipelines: Reference architecture

1. A data lakehouse architecture typically employs three layers: Land, Structure, and Consume.
2. The Land Layer receives raw data.
3. It is then transformed to create the Structure layer.
4. Finally, additional transformations occur to create the Consume layer, which are the datasets queried by dashboards and users.
5. This entire process is ETL, and a number of different tools can be used to enable it.